

Assignment 2 (midterm). DTU-Autumn 2026

02429 Analysis of correlated data: Linear mixed models

October 1, 2026

To be uploaded to dtulearn by **Monday the 12th by 9:00**. And solved in groups of 3 students, groups of size 1 are NOT accepted.

The report is **roughly** composed as follows:

- i) A **maximum of 12 pages** for the main text.
- ii) Include as many appendices as necessary for additional graphs and tables of interest.
- iii) Include one appendix with the output for

- `ranova(final_model)`
- `drop1(final_model, test="F")`
- `summary(final_model)`, if too long, include an abbreviated version (fewer rows)

These will allow us to verify the numerical results.

- iv) An appendix with the full R code at the end of the report.
- v) Upload a **single pdf** file with name analogous to: **Assignment2_Group1.pdf**

Remember: use no more than four **significant digits** and fewer if possible. Use a good font size for numbers and labels in the figures. All figures and tables must be referred to in the text, should be numbered and have an **informative legend**. Balance the length of the different sections (exploratory analysis, modelling, ..., conclusions) in the main text. When writing your solutions consider that **the readers** are the class members. Use **font size** 11 or 12 for the main text.

The solution of each exercise can **roughly** follow the following steps.

I. Exploratory analysis.

- Do a graphical and numerical exploratory analysis. What do you see?

II. Modelling.

- Start with a model and **build up or down** a potential final model.
- Present **diagnostic** plots/statistics of the final model.
- Give a **precise algebraic expression**, numbered, of the final proposed model.

III Results.

- Give the numerical results that answer the questions.
- When presenting parameter estimates remember to include confidence intervals.
- Present any additional results you may consider relevant or informative.

Exercise I. Ca concentration in turnip plants. A components of variance model.

Consider a sample of measurements with a hierarchical structure. It is a random sample of two Calcium measurements on each of three randomly selected leaves from each of four randomly selected plants. The data are shown in the table below. No data file is provided.

In this example, the process of sampling and subsampling gives rise to nested or hierarchical classifications. And the total sample of observations could be modelled with a linear mixed model with nested factors.

TABLE 13.8.1
CALCIUM CONCENTRATION (% DRY BASIS) IN $b = 3$ LEAVES FROM EACH OF $a = 4$ TURNIP PLANTS, $n = 2$ DETERMINATIONS/LEAF. ANALYSIS OF VARIANCE

Plant, i $i = 1, \dots, a$	Leaf, ij $j = 1, \dots, b$	Determinations, X_{ijk}		$X_{ij.}$	$X_{i..}$	$X_{...}$			
1	1	3.28	3.09	6.37	19.05	72.29			
	2	3.52	3.48	7.00					
	3	2.88	2.80	5.68					
2	1	2.46	2.44	4.90	13.07		72.29		
	2	1.87	1.92	3.79					
	3	2.19	2.19	4.38					
3	1	2.77	2.66	5.43	17.71			72.29	
	2	3.74	3.44	7.18					
	3	2.55	2.55	5.10					
4	1	3.78	3.87	7.65	22.46				72.29
	2	4.07	4.12	8.19					
	3	3.31	3.31	6.62					
total size = $abn = (4)(3)(2) = 24$ determinations									

Snedecor & Cochran, Statistical Methods, p 247.

Exercise:

- i) Fit a linear mixed model and estimate the components of variance.
- ii) Suppose that the researcher wished to repeat the experiment with double the size of the sample. What do you advise to increase: the number of plants, leaves or measurements?
- iii) What is the estimate for the mean value of calcium concentration? Give confidence intervals calculated in three ways: a) using the standard error provided in the summary of the fitted model, b) the profile CI, and c) CI computed using bootstrap.
- iv) Now, estimate the mean value of calcium concentration ignoring the hierarchical structure using a linear model. Compare the results with those in iii). Present a table or figure and comment.

Exercise II. Reproductive study example.

'In reproductive studies on rats to assess the effects of an experimental compound on general reproductive performance and postnatal measurements taken on offspring (pups), pup weights are used as one indicator of treatment effects. For each pup, there are two sources of variation: the mother (dam) and the within-litter variance. This suggests the use of a mixed model with two variance components. Additionally, there are other uncontrollable factors which may also influence pup weights. Pups from larger litters tend to weigh less than those from litters with fewer pups, and males generally weigh more than females. Both litter size and sex are used as covariates in the analysis'.

TABLE 1
Pup weights (g) in a reproductive study

<i>Dose group</i>	<i>Dam</i>	<i>Males</i>	<i>Females</i>
Controls	1	6.60, 7.40, 7.15, 7.24, 7.10, 6.04, 6.98, 7.05	6.95, 6.29, 6.77, 6.57
	2	6.37, 6.37, 6.90, 6.34, 6.50, 6.10, 6.44, 6.94, 6.41	5.92, 6.04, 5.82, 6.04, 5.96
	3	7.50, 7.08	7.57, 7.27
	4	6.25, 6.93, 6.80, 6.69, 6.28, 6.27, 6.27, 6.47	6.29, 5.98, 6.32, 6.28, 5.65, 5.57
	5	7.96, 6.84, 7.00, 8.10, 6.52, 7.23, 6.10, 7.31	7.16, 7.09, 7.14, 5.02, 6.04
	6	8.26, 7.73, 8.33, 6.14, 7.75, 6.96	7.26, 6.58, 3.68
	7	6.29, 6.32, 6.28, 6.24, 6.78, 6.63, 6.27, 6.29, 6.06	6.16, 5.96, 6.26, 5.83, 6.11, 6.45, 6.25, 6.31, 5.74
	8	6.04, 5.84, 6.77, 5.59, 5.52, 6.42, 5.97, 6.34	6.23, 5.95, 6.16, 6.19, 5.32, 5.00, 6.30, 5.00, 5.56
	9	5.37, 5.58, 5.51, 5.19, 5.34, 5.77, 5.17, 4.57, 5.39, 5.62, 5.40, 5.77, 5.24	5.37, 5.33, 5.44, 5.14
	10	7.30, 6.60, 6.58, 6.68, 6.46, 6.38	6.44, 6.67, 6.43, 6.53, 5.92, 6.52, 6.44
Low dose	11	5.65, 5.78, 6.23, 5.70, 5.73, 6.10, 5.55, 5.71, 5.81, 6.10	5.54, 5.72, 5.50, 5.64, 5.42, 5.42
	12		6.89, 7.73
	13	5.83, 5.97, 6.39, 5.69, 5.69, 5.97, 6.04, 5.46	6.09, 5.39, 5.89, 5.14
	14	5.92, 5.75, 6.22, 5.96, 5.59, 5.79, 6.23, 5.88, 6.02	5.66, 5.76, 5.73, 5.33, 5.58, 5.88
	15	6.00, 6.11, 6.40, 6.06, 6.39, 6.09, 6.32	5.96, 6.32, 5.83, 5.97, 5.87, 5.67
	16	6.43, 6.13, 5.87	6.09, 5.63, 5.84, 6.20, 6.42, 5.90, 5.62, 6.23, 5.85, 5.89
	17	5.81, 5.44, 5.65, 5.25, 5.45, 5.32, 5.89	5.63, 5.12, 5.65, 5.29, 5.13, 5.60, 5.08
	18	6.77, 7.13, 6.85	6.49, 6.09, 6.09, 5.99, 6.01, 6.11, 6.15, 4.75, 5.69, 6.19, 5.72, 6.14
	19	6.72, 6.34, 6.48, 5.74	6.11, 5.71, 6.41, 6.21, 6.11, 5.81
	20	5.90, 6.22, 6.67, 6.23, 6.24, 6.26, 6.38, 6.05, 5.89, 6.29	6.12, 5.40, 5.50, 5.46, 5.97, 6.11
High dose	21	5.09, 5.57, 5.69, 5.50, 5.45, 5.24, 5.36, 5.26, 5.36, 5.01, 5.03	5.23, 5.13, 4.48
	22	5.30, 5.40, 5.55, 6.02, 5.27	5.19, 5.42, 5.40, 5.12, 5.40
	23	7.70	7.68, 6.33
	24	6.28, 5.74, 6.29	5.68, 5.76, 6.03, 5.30, 5.55, 6.53, 5.76, 5.77, 5.49
	25	6.50, 7.10, 7.00, 7.00, 5.85	6.10, 6.63, 6.33
	26	7.00, 6.15	6.22, 6.20, 5.76, 6.21, 6.42, 6.42, 6.30
	27	5.64, 6.06, 6.56, 6.29, 5.69, 6.36	5.93, 5.74, 5.74

Table from A.P. Dempster et al.. Appl. Statist (1984). `Pulps_Dempster.RData`

Exercise: Analyze the data in the table in order to assess the effects of the treatment, use a linear mixed model. Do a full statistical analysis to answer the main research question, and present relevant information and findings. All findings should be accompanied by sound statistical results.

Exercise III. Simulation study

In Exercise I, Assignment 1, we used an example of a simple random effects model

$$y_{ij} = \mu + a_i + \varepsilon_{ij}, \quad (1)$$

$$a_i \sim N(0, \sigma_a^2), \quad \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2), \quad \text{Cov}(a_i, \varepsilon_{ij}) = 0, \quad i = 1 \dots, G, \quad j = 1, \dots, n_i.$$

i) Simulate data from the model with groups of different sizes and a number of groups of your choice. Consider the following six scenarios:

- a) two different total sample sizes $N = \sum_i n_i$: small/ large, and
- b) three intraclass correlation values: 0, .45, .95.

Report in tabular form, say in Table 1, the estimated mean value with CI, the components of variance, the estimated intraclass correlation value, and the corresponding population/theoretical parameter values. Include some plots of the data. More than one plot in a figure for easy reading. Compare the results obtained in Table 1, and comment. Use auxiliary plots.

ii) [Optional] Repeat $B = 100$ times the simulation in i). Calculate the **average** of the estimated mean values and its variance, and produce its CI. Include your results in Table 1. Compare the results obtained in Table 1, and comment. Use auxiliary plots.